

# Distributed Topological Charge and Spinorial Holonomy in Yukawa-Regularized Graphene Disclinations

A. M. de M. Carvalho<sup>1,\*</sup>, G. Q. Garcia<sup>2,†</sup>, E. Brito<sup>3,4,‡</sup> and C. Furtado<sup>4,§</sup>

<sup>1</sup>*Instituto de Física, Universidade Federal da Alagoas, 57072-970, Maceió, AL, Brazil.*

<sup>2</sup>*Centro de Ciências, Tecnologia e Saúde, Universidade Estadual da Paraíba, 58233-000, Araruna, PB, Brazil.*

<sup>3</sup>*Centro de Ciências Exatas e das Tecnologias, Universidade Federal do Oeste da Bahia, 47810-059, Barreiras, BA, Brazil*

<sup>4</sup>*Departamento de Física, Universidade Federal da Paraíba, 58051-970, João Pessoa, PB, Brazil.*

Conical geometries provide the standard description of disclinations, but they concentrate the curvature at a singular apex. We introduce a Yukawa-type regularization that replaces this singularity by a smooth curvature distribution while preserving the asymptotic topology of the defect. Exact expressions are obtained for the conformal factor, curvature, and enclosed topological charge. The resulting geometry exhibits a scale-dependent topological charge and a corresponding radius-dependent holonomy, establishing a direct connection between distributed curvature and geometric phases. We further investigate the dynamics of massless Dirac quasiparticles in this background and show that the regularized core modifies the spin connection while preserving the asymptotic topological signature of the defect. These results provide a finite-core extension of the conventional conical description and offer a natural framework for studying geometric and topological effects in graphene-like systems.

PACS numbers: 04.50.Kd, 61.72.Lk, 02.40.Ky, 04.20.Jb

## I. INTRODUCTION

The geometric interpretation of topological defects has a long history, beginning with the pioneering works of Kondo in ref. [1], Bilby *et. al* in ref. [2], and later developed systematically by Kröner [3] and Kleinert [4]. These ideas culminated in the geometric theory of defects [5] formulated by Katanaev and Volovich, in which disclinations and dislocations are associated with curvature and torsion, respectively. This framework establishes a direct connection between the theory of defects in condensed matter and differential geometry, describing elastic and topological properties in terms of geometric fields [6–8]. Within this approach, disclinations are commonly described by conical geometries whose curvature is concentrated at a singular apex, reproducing the characteristic holonomy associated with the defect while preserving the local flatness of the surrounding space [9–11]. And although ideal conical defects successfully capture the global topological features of the geometry, their curvature is represented by a distributional source localized at a single point.

While this approximation correctly reproduces the asymptotic topology, it neglects the finite spatial extent expected for realistic defect cores. As a result, geometric invariants become singular at the apex, and the description of short-distance phenomena, including the interaction of quantum particles with the defect core, becomes less transparent. A natural regularization consists of replacing the singular source with a continuous curvature distribution that smooths the core while preserving the asymptotic topological structure of the defect. Recent developments have explored this idea within the conformal metric approach, where the geometry is generated by continuous distributions of disclination-like defects. In this framework, Gaussian, exponential, and other smooth curvature profiles lead to regular geometries with finite curvature and well-defined topological charges, providing a continuous generalization of ideal conical defects. These studies showed that the conformal factor can be obtained from a Poisson equation sourced by the defect distribution, allowing the construction of exact finite-core solutions and the investigation of the relation between local curvature and global topology [10].

Unlike these regularization schemes, the Yukawa kernel arises naturally as the Green's function of the two-dimensional modified Helmholtz operator, introducing an intrinsic screening length into the geometry. Originally proposed to describe screened interactions, the Yukawa potential has been widely used to model complex plasmas, colloidal suspensions, and other strongly correlated systems, where it reproduces structural, thermodynamic, and defect-related properties over a broad range of physical conditions [12, 13]. When employed as a curvature source, the Yukawa kernel replaces the singular apex of an ideal conical defect by a smooth finite-core distribution while

\* alexandre@fis.ufal.br

† ggarcia99@gmail.com

‡ eliasbaj@ufob.edu.br

§ furtado@fisica.ufpb.br

preserving the asymptotic conical geometry. This combination of analytical tractability and an intrinsic geometric length scale makes it particularly well suited for investigating how local regularization influences global topological properties and the associated holonomy. While smooth defect geometries have been investigated in several contexts, considerably less attention has been paid to how the associated curvature and holonomy are distributed throughout space. In ideal conical geometries, the entire topological content is concentrated at the apex, and the holonomy is fully determined by a single deficit angle [5, 9, 14]. For smooth curvature distributions, however, both quantities become scale dependent, revealing a richer interplay between local geometry and global topology. This observation motivates the central question addressed in this work: how does a finite-core regularization modify the spatial accumulation of curvature and the resulting geometric phase?

In this paper, we construct an exact conformal geometry generated by a Yukawa-type curvature distribution and investigate its geometric, topological, and spinorial properties. The resulting solution possesses a regular core and approaches the standard conical geometry asymptotically. Closed-form expressions are obtained for the conformal factor, curvature, and enclosed topological charge. We show that the topological charge becomes radius-dependent and is recovered only asymptotically, leading naturally to a scale-dependent holonomy that interpolates between locally Euclidean behavior and the asymptotic conical regime.

This resulting geometry allows us to investigate the dynamics of massless Dirac quasiparticles. In the continuum description of graphene, low-energy electronic excitations behave as relativistic fermions coupled to effective gauge fields and curved backgrounds [15–19], since that the graphene is a kind of material closely associated with topological defects. Indeed, graphene is a crystal lattice composed of carbon atoms in a honeycomb form, and it is virtually impossible to obtain it without the presence of topological defects. Topological defects in graphene can be described by the Volterra processes [20], in which an angular sector multiple of  $\frac{\pi}{3}$  can be added and/or removed from a flat graphene. As a consequence, disclinations arise due to the presence of pentagonal or heptagonal rings and dislocations due to the presence of pentagon-heptagon pairs. In the low-energy limit, graphene can be described by the massless Dirac equation, and the topological defects act directly in the parallel quantum transportation of graphene quasiparticles. We also extend the analysis to massless Dirac quasiparticles propagating in the regularized geometry and show that the finite curvature distribution modifies the spin connection while preserving the asymptotic spinorial holonomy associated with the defect.

The present construction extends previous analyses based on Gaussian and exponential curvature distributions by introducing a finite screening length as an independent geometric scale. It replaces the singular apex of an ideal conical defect with a finite core while preserving its asymptotic topological structure. As a consequence, the enclosed topological charge, the associated holonomy, and the corresponding geometric phase become scale-dependent, reflecting the progressive accumulation of curvature with distance. The same framework also allows us to investigate the dynamics of massless Dirac quasiparticles in graphene-like systems and the interplay between local curvature and global topological properties. This work is organized as follows. Section II introduces the geometric framework. Section III presents the Yukawa-regularized geometry. Sections IV and V analyze the curvature, topological charge, and scale-dependent holonomy. Section VI investigates the dynamics of massless Dirac quasiparticles in the regularized geometry. Finally, Section VII presents the conclusions.

## II. GEOMETRIC FRAMEWORK

We adopt the geometric theory of defects developed by Katanaev and Volovich, in which curvature describes disclinations and torsion describes dislocations. This approach establishes a direct correspondence between defect theory and Riemann–Cartan geometry, allowing the metric and connection to encode the elastic and topological properties of the medium [4–7]. For an infinite and static defect line, the medium is invariant under translations along the defect axis. Consequently, the physical properties of the defect are completely determined by the geometry of the transverse two-dimensional section. Within the geometric theory of defects, disclinations are associated with curvature, whereas dislocations are associated with torsion. The topological character of the defect is encoded in the behavior of closed curves surrounding the defect core, giving rise to quantities such as the deficit angle and the corresponding holonomy. Since any two-dimensional Riemannian geometry is locally conformally flat, the transverse metric can be written in conformal form. The metric therefore takes the form

$$ds^2 = e^{2\Omega(r)}(dr^2 + r^2 d\theta^2) + dz^2, \quad (2.1)$$

where  $\Omega = \Omega(r)$  is the conformal factor, depending only on the radial coordinate. The conformal factor fully determines the geometry of the defect. Once  $\Omega$  is known, the curvature, topological charge, and holonomy follow directly from the resulting geometric structure.

The Einstein-like field equation, expressed as

$$R_{\alpha\beta} - \frac{1}{2}g_{\alpha\beta}R = -8\pi GT_{\alpha\beta}, \quad (2.2)$$

where  $R_{\alpha\beta}$  and  $R$  denote the Ricci tensor and Ricci scalar, respectively. The tensor  $T_{\alpha\beta}$  represents the defect density and acts as the source of the stress and strain fields. The parameter  $G$  plays the role of a coupling constant and is associated with the elastic properties of the continuum medium. In the geometric theory of defects, the continuum is modeled as a Riemann–Cartan medium, and the metric–connection pair satisfies an Einstein-type field equation with defect densities as sources [21]. Disclinations source curvature and dislocations source torsion; away from cores, the geometry is locally Euclidean. In the translationally invariant setting described by (2.1), the Einstein-like equation (2.2) reduces to a Poisson-like equation for the conformal factor [22]

$$\Delta\Omega(r) = -\lambda(r), \quad (2.3)$$

Here  $\lambda(r)$  denotes the prescribed curvature distribution in the transverse plane [5].

### III. YUKAWA-SCREENED CURVATURE

We prescribe the curvature density by the two-dimensional modified Helmholtz kernel  $K_0(\mu r)$  [23, 24], given by

$$\lambda(r) = \frac{\Lambda\mu^2}{2\pi} K_0(\mu r), \quad \mu > 0, \quad (3.1)$$

normalized so that

$$2\pi \int_0^\infty r \lambda \, dr = \Lambda. \quad (3.2)$$

The normalization ensures that the total integrated Gaussian curvature is fixed at  $\Lambda$ , independently of the screening parameter  $\mu$ . Thus,  $\Lambda$  characterizes the overall strength of the curvature distribution, while  $\mu^{-1}$  sets the corresponding screening length [25]. This choice is motivated by three main properties of the Yukawa kernel: (i) finite-range screening, which localizes the curvature near the defect core; (ii) compatibility with the logarithmic behavior characteristic of conical geometries at short distances; and (iii) analytical tractability, since the resulting conformal factor can be obtained in closed form and yields stable geometric observables. The small- and large-argument asymptotics of  $K_0(\mu r)$  (with  $\ell = \mu^{-1}$ ) are [26]:

$$K_0(\mu r) = -\ln\left(\frac{\mu r}{2}\right) - \gamma + \mathcal{O}((\mu r)^2 \ln(\mu r)) \quad (\mu r \rightarrow 0), \quad (3.3)$$

and

$$K_0(\mu r) = \sqrt{\frac{\pi}{2\mu r}} e^{-\mu r} \left[ 1 + \mathcal{O}\left(\frac{1}{\mu r}\right) \right] \quad (\mu r \rightarrow \infty). \quad (3.4)$$

The asymptotic behavior of the Yukawa kernel ensures a localized core with an exponentially decaying tail and a smooth crossover to the asymptotic conical geometry while preserving the global deficit determined by  $\Lambda$ .

We now derive the explicit form of the conformal factor  $\Omega(r)$ , which determines the metric generated by the prescribed curvature distribution. The solution of (2.3) with curvature distribution (3.1) is unique up to an additive constant

$$\Omega(r) = -\frac{\Lambda}{2\pi} \ln r - \frac{\Lambda}{2\pi} K_0(\mu r) + C. \quad (3.5)$$

Imposing core regularity and fixing the asymptotic deficit determine the additive constant. We fix the gauge by enforcing a regular apex,

$$\Omega(0) = 0 \quad \text{and} \quad \Omega'(0) = 0, \quad (3.6)$$

which removes spurious constants and makes the core value of  $\Omega$  directly interpretable. Accordingly, using the small-argument expansion (3.3), we choose the additive constant as

$$C = -\frac{\Lambda}{2\pi} \left( \ln \frac{\mu}{2} + \gamma \right), \quad (3.7)$$

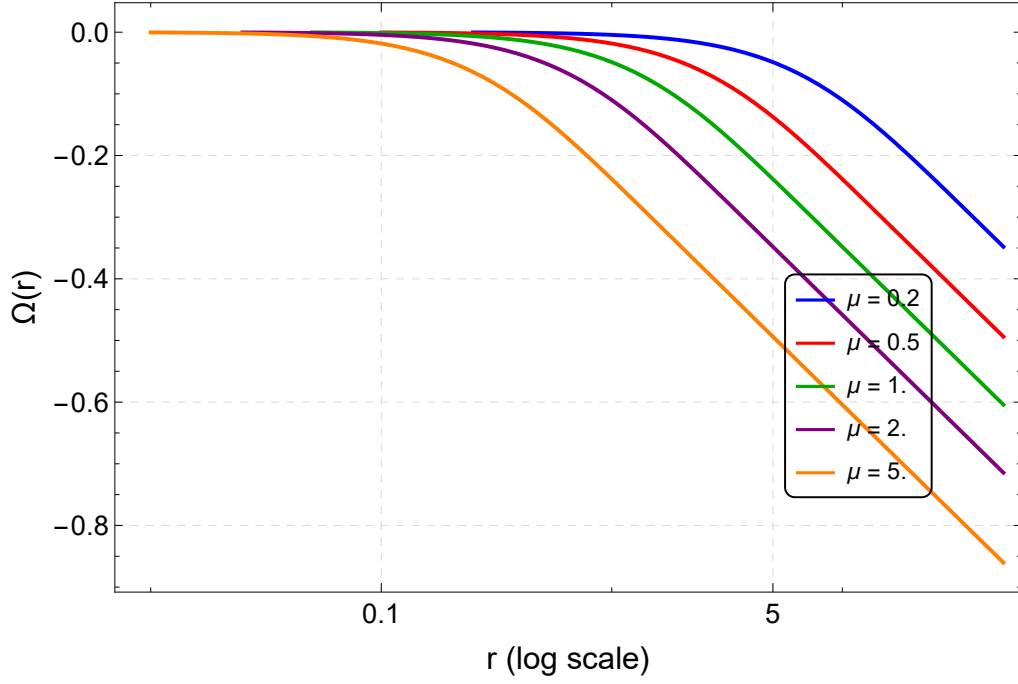


FIG. 1. Conformal factor  $\Omega(r)$  (core-regular gauge,  $\Omega(0) = 0$ ) as a function of  $r$  (logarithmic scale) for several values of the regularization parameter  $\mu$ .

so that  $\Omega(r) \rightarrow 0$  as  $r \rightarrow 0$ . To ensure a regular apex and physically meaningful geometry, we impose the boundary condition

$$\Omega(0) = 0, \quad (3.8)$$

which guarantees that the metric is locally Euclidean at the core and removes spurious constants, in accordance with the geometric regularity criteria.

The final expression for the conformal factor is given by

$$\Omega(r) = -\frac{\Lambda}{2\pi} \left[ \ln\left(\frac{\mu r}{2}\right) + \gamma + K_0(\mu r) \right], \quad (3.9)$$

so that the induced metric reads

$$ds^2 = e^{2C} r^{-\Lambda/\pi} \exp\left[-\frac{\Lambda}{\pi} K_0(\mu r)\right] (dr^2 + r^2 d\theta^2). \quad (3.10)$$

Substituting (3.7), the metric (3.10) becomes

$$ds^2 = \left(\frac{\mu}{2}\right)^{-\Lambda/\pi} e^{-(\Lambda/\pi)\gamma} r^{-\Lambda/\pi} \exp\left[-\frac{\Lambda}{\pi} K_0(\mu r)\right] (dr^2 + r^2 d\theta^2). \quad (3.11)$$

The metric (3.11) describes a screened disclination: the Yukawa profile smooths the apex, yielding a finite, locally Euclidean core, while the far field remains asymptotically conical [9, 14, 27]. The crossover is set by the screening length  $\ell = \mu^{-1}$ , so the geometry interpolates smoothly from the regular core to an asymptotic cone whose deficit angle is determined by the parameter  $\Lambda$ . As shown in Fig. 1, the conformal factor remains essentially constant in the core region (small  $r$ ), reflecting the regularization that removes the apex singularity. As  $r$  increases, a crossover takes place around the characteristic scale  $r \sim 1/\mu$ , beyond which  $\Omega(r)$  departs from the core plateau and approaches the far-field behavior governed by the logarithmic term. Increasing  $\mu$  shrinks the effective core and shifts the crossover to smaller radii, making explicit how  $\mu$  controls the size of the regularized region.

### A. Asymptotic Analysis

We analyze the asymptotic behavior of the metric. First, consider the near-core limit  $r \rightarrow 0$ . Using (3.3), the conformal factor behaves as

$$\Omega(r) = \mathcal{O}((\mu r)^2 \ln r). \quad (3.12)$$

Consequently,

$$e^{2\Omega(r)} = 1 + \mathcal{O}((\mu r)^2 \ln r). \quad (3.13)$$

Hence the line element approaches the Euclidean form,

$$ds^2 \sim dr^2 + r^2 d\theta^2 \quad (3.14)$$

with only subleading corrections.

We now study the far-field limit ( $r \rightarrow \infty$ ), corresponding to Eq. (3.4). To cast the metric in standard conical form, introduce the proper radial variable

$$\rho = \frac{e^C}{1 - \Lambda/2\pi} r^{1-\Lambda/2\pi}, \quad (\Lambda \neq 2\pi). \quad (3.15)$$

With this change of variable, the metric (3.11) tends to the conical form

$$ds^2 \sim d\rho^2 + \alpha^2 \rho^2 d\theta^2, \quad (3.16)$$

where  $\alpha$  is the conventional disclination parameter that characterizes the conical geometry (and hence the deficit angle),

$$\alpha = 1 - \frac{\Lambda}{2\pi}. \quad (3.17)$$

The deficit/excess angle is given by

$$\delta = \Lambda = 2\pi(1 - \alpha). \quad (3.18)$$

The azimuthal angle ranges over  $(0, 2\pi\alpha)$ . For  $\alpha < 1$ , the geometry corresponds to a deficit angle, whereas  $\alpha > 1$  describes an excess angle. Equivalently,  $\delta > 0$  corresponds to a deficit angle and  $\delta < 0$  to an excess angle.

## IV. CURVATURE DISTRIBUTION, EFFECTIVE CHARGE, AND GAUSS–BONNET

To characterize the curvature of the regularized geometry, we compute the Ricci scalar. For the conformal metric (2.1), one finds

$$R = -2 e^{-2\Omega} \nabla^2 \Omega. \quad (4.1)$$

For the specific case of the conformal factor (3.9),

$$R(r) = \frac{\Lambda}{\pi} \left( \frac{\mu}{2} \right)^{\Lambda/\pi} \mu^2 e^{(\Lambda/\pi)\gamma} r^{\Lambda/\pi} \exp \left[ \frac{\Lambda}{\pi} K_0(\mu r) \right] K_0(\mu r). \quad (4.2)$$

For  $r \rightarrow 0$ , the curvature does not vanish. Instead, it develops an integrable core peak associated with the logarithmic behavior of the modified Bessel function  $K_0(\mu r)$ . Although  $K_0$  diverges logarithmically at the origin, the corresponding curvature profile remains locally integrable, reflecting a smooth but sharply localized core structure. The condition  $\Omega(0) = 0$  ensures regularity of the metric at the apex, but it does not imply  $R(0) = 0$ . Rather, the curvature is concentrated within a finite region whose characteristic size is  $\ell \sim \mu^{-1}$ . For  $r \rightarrow \infty$ , the curvature decays exponentially, exhibiting a Yukawa-type screening. The conical limit is recovered by shrinking the range of the curvature profile, namely by taking  $\mu \rightarrow \infty$  while keeping the total integrated Gaussian curvature fixed

$$\int K dA = \Lambda. \quad (4.3)$$

This equation directly reflects the normalization condition (3.2), showing that the total integrated curvature of the geometry is entirely determined by the prescribed source distribution.

In this limit, the smooth distribution collapses to a Dirac delta distribution,

$$K(r) \longrightarrow \Lambda \delta^{(2)}(r), \quad (4.4)$$

reproducing the ideal conical geometry, for which  $K = 0$  away from the apex and the total curvature is concentrated at the tip. The curvature enclosed within a finite radius becomes a scale-dependent quantity. Defining the effective enclosed topological charge by

$$Q(\rho) = \int_0^\rho K \sqrt{g} d^2x, \quad (4.5)$$

one finds that  $Q(\rho)$  increases continuously with the radial coordinate and approaches the total charge only asymptotically,

$$\lim_{\rho \rightarrow \infty} Q(\rho) = \Lambda. \quad (4.6)$$

Within this framework, the Gauss-Bonnet theorem [28] provides the corresponding global constraint. For an annulus  $\mathcal{A}_{\epsilon,R}$  with smooth circular boundaries, one has

$$\int_{\mathcal{A}_{\epsilon,\rho}} K \sqrt{g} d^2x + \oint_{\partial \mathcal{A}_{\epsilon,\rho}} k_g ds = 0, \quad (4.7)$$

since  $\chi(\mathcal{A}_{\epsilon,\rho}) = 0$ . Taking the limits  $\epsilon \rightarrow 0^+$  and  $\rho \rightarrow \infty$ , the oriented boundary terms reproduce the conical contributions: the inner circle yields  $-2\pi$ , whereas the outer circle tends to  $+2\pi\alpha$ . Therefore,

$$\int_{\mathcal{A}_{0,\infty}} K \sqrt{g} d^2x = 2\pi(1 - \alpha) = \Lambda. \quad (4.8)$$

Consequently, the total integrated Gaussian curvature, and hence the topological charge, remains fixed and is determined by the asymptotic boundary contribution. The Gauss-Bonnet theorem ensures that the global topology depends only on the integrated curvature, not on its specific spatial distribution. Thus, although the regularization spreads the curvature over a finite region, the asymptotic deficit angle and the corresponding holonomy remain unchanged. Any loop enclosing the defect at a sufficiently large radius accumulates the same rotation angle  $\delta = \Lambda$ . The figure 2 shows that the curvature is localized near the core and rapidly suppressed away from the origin, consistent with a Yukawa-type regularization. The parameter  $\mu$  sets the characteristic core length  $r \sim 1/\mu$ : increasing  $\mu$  narrows the curvature peak and shifts the decay to smaller radii, while smaller  $\mu$  spreads the curvature over a wider region. This makes explicit how  $\mu$  controls the spatial extent of the regularized defect, while preserving a well-defined far-field regime.

## V. HOLONOMY AND EFFECTIVE TOPOLOGICAL CHARGE

A natural way to characterize the observable geometric effects of a topological defect is through its holonomy. In differential geometry, the holonomy measures the net rotation acquired by a vector, spinor, or tensor after parallel transport around a closed curve [29–33]. For a loop  $\gamma$ , the corresponding holonomy transformation is defined by the path-ordered exponential

$$H(\gamma) = P \exp \left( - \oint_\gamma \Gamma_\mu(x) dx^\mu \right), \quad (5.1)$$

where  $\Gamma_\mu$  denotes the connection and  $P$  represents path ordering. The holonomy provides a direct probe of the curvature enclosed by the loop. In particular, for conical geometries associated with disclinations, it encodes the deficit angle and therefore the global topological content of the defect. This geometric phase is closely analogous to the phase factor appearing in the Aharonov-Bohm effect, where the observable quantity depends on the flux enclosed by the trajectory rather than on local fields along the path. For the Yukawa-regularized geometry, the curvature is not concentrated at a singular apex but distributed over a finite region. Consequently, the holonomy becomes

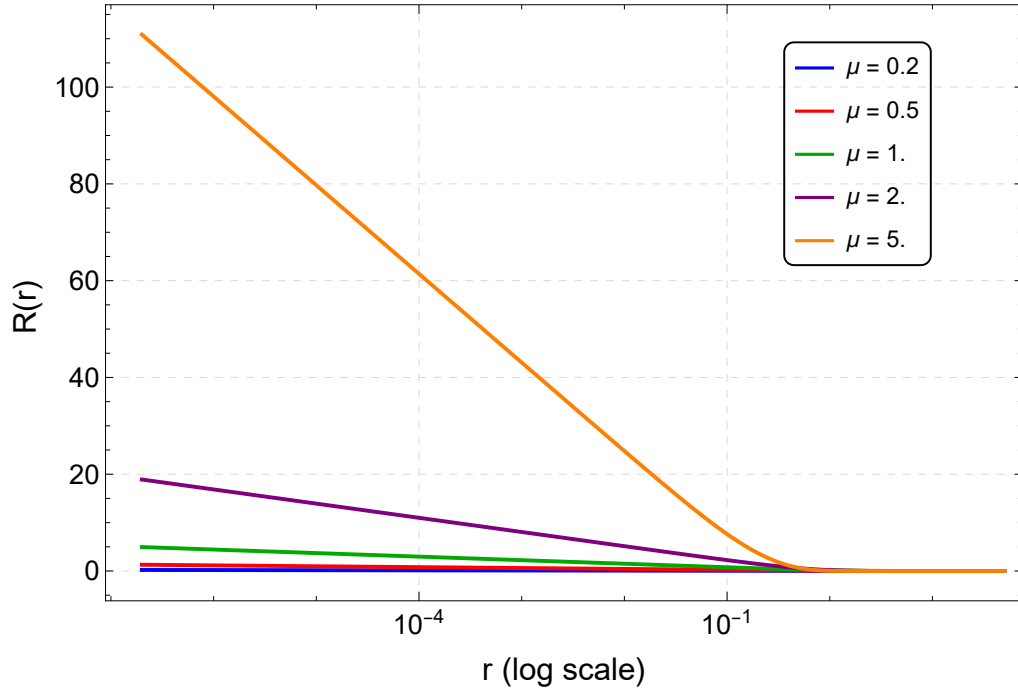


FIG. 2. Scalar curvature  $R(r)$  given by Eq. (4.2) as a function of the radial coordinate  $r$  (logarithmic scale) for the same set of  $\mu$  values used in Fig. 1.

radius-dependent. A circular loop of radius  $\rho$  encloses only a fraction of the total curvature, quantified by the effective charge

$$Q(\rho) = \int_0^\rho K \sqrt{g} d^2x. \quad (5.2)$$

The associated holonomy therefore evolves continuously with the loop radius and approaches the conical result only asymptotically.

The same quantity can be related to the deficit angle obtained from the holonomy of a circular path. Following Ref. [11], for a conformal metric, the angle associated with parallel transport around a circle of radius  $\rho$  is

$$\delta(\rho) = - \int_0^{2\pi} \rho \frac{\partial \Omega}{\partial r} \Big|_{r=\rho} d\theta. \quad (5.3)$$

Since the conformal factor is radial, this reduces to

$$\delta(\rho) = -2\pi\rho\Omega'(\rho). \quad (5.4)$$

Substituting the Yukawa-regularized conformal factor,

$$\Omega'_Y(\rho) = -\frac{\Lambda}{2\pi\rho} + \frac{\Lambda\mu}{2\pi}K_1(\mu\rho), \quad (5.5)$$

one obtains

$$\delta(\rho) = \Lambda[1 - \mu\rho K_1(\mu\rho)]. \quad (5.6)$$

This expression coincides exactly with the effective topological charge enclosed by a circle of radius  $\rho$ , previously obtained from the integrated curvature. Since the parameter  $\rho$  simultaneously labels the transport loop and the region enclosed by it, the holonomy provides a direct measure of the accumulated curvature,

$$Q(\rho) = \Lambda[1 - \mu\rho K_1(\mu\rho)]. \quad (5.7)$$



Therefore,

$$\delta(\rho) = Q(\rho), \quad (5.8)$$

showing that the deficit angle measured by parallel transport is precisely determined by the curvature enclosed by the transport loop. The limiting behaviors follow immediately:

$$\lim_{\rho \rightarrow 0} \delta(\rho) = 0, \quad \lim_{\rho \rightarrow \infty} \delta(\rho) = \Lambda. \quad (5.9)$$

From a geometric perspective, the Yukawa regularization replaces the abrupt topological imprint of an ideal disclination by a smooth radial accumulation of holonomy. The enclosed topological charge, the deficit angle and the holonomy become scale-dependent quantities that interpolate continuously between the regular core and the asymptotic cone. This behavior provides a natural bridge between the local curvature distribution and the global topological properties of the defect. A discrete holonomy invariance may also occur for special circular loops, following the mechanism first discussed by Rothman *et al.* for Schwarzschild geometry [34]. Since the holonomy associated with a circular loop corresponds to a rotation by the angle  $\delta(\rho)$ , after  $n$  successive windings the accumulated rotation is  $n\delta(\rho)$ . The holonomy therefore becomes trivial whenever

$$n\delta(\rho) = 2\pi m, \quad m \in \mathbb{Z}. \quad (5.10)$$

Using the identity  $\delta(\rho) = Q(\rho)$  together with Eq. (5.7), this condition becomes

$$n\Lambda [1 - \mu\rho K_1(\mu\rho)] = 2\pi m. \quad (5.11)$$

Hence, the critical radii are determined by the enclosed topological charge rather than directly by the metric coefficients. The figure 3 illustrates the scale dependence of the enclosed topological charge. The transition from  $Q(\rho) = 0$  near the regular core to  $Q(\rho) = \Lambda$  in the asymptotic region reflects the gradual accumulation of curvature and, consequently, of the associated holonomy. As discussed above, the discrete holonomy-invariance condition is satisfied when the enclosed charge reaches specific fractions of the total charge, thereby defining the corresponding critical radii. The horizontal dashed lines indicate representative values of these fractions. Increasing  $\mu$  concentrates the curvature within a smaller region, causing the enclosed topological charge to saturate more rapidly and shifting the critical radii toward smaller values. Thus, the holonomy vanishes at the regular core and approaches the asymptotic conical deficit angle only at large distances.

## VI. MASSLESS DIRAC FERMIONS IN THE YUKAWA-REGULARIZED GEOMETRY

One of the most remarkable properties of graphene is that its low-energy electronic excitations are not described by the Schrödinger equation, but by a two-dimensional massless Dirac equation. Near the inequivalent  $K$  and  $K'$  points of the Brillouin zone, the electronic spectrum becomes linear, and the quasiparticles behave as relativistic fermions with the Fermi velocity  $v_F$  playing the role of an effective speed of light [15–17]. Within the continuum description of graphene, elastic deformations, topological defects, and curvature can be incorporated geometrically through effective gauge fields and curved metrics acting on the Dirac quasiparticles [17, 18]. In this framework, disclinations are naturally associated with conical geometries, while smooth curvature distributions may be interpreted as regularized defect cores [18, 35].

The Yukawa-regularized geometry developed in the previous sections provides a finite-core extension of the conventional conical description of graphene disclinations. Instead of concentrating the curvature at a singular apex, the geometry distributes it over a finite region characterized by the length scale  $\mu^{-1}$ . As discussed in the previous sections, this regularization produces a radius-dependent topological charge and a corresponding scale-dependent holonomy. It is therefore natural to investigate how massless Dirac quasiparticles respond not only to the total topological charge of the defect but also to the detailed structure of its geometric core.

### A. Dirac equation in the conformal geometry

The dynamics of massless Dirac quasiparticles in the conformal background (2.1) is described by the curved-space Dirac equation [15, 18],

$$i\gamma^\mu(x)\nabla_\mu\Psi = 0, \quad (6.1)$$



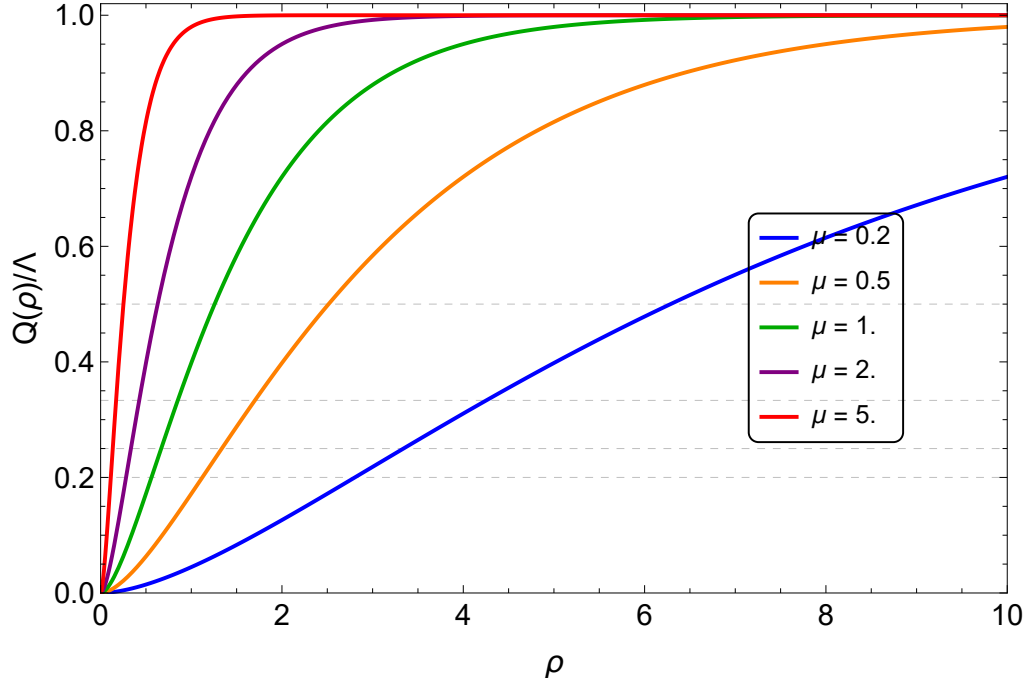


FIG. 3. Normalized enclosed topological charge  $Q(\rho)/\Lambda$  as a function of the radial coordinate  $\rho$  for different values of the screening parameter  $\mu$ . The charge increases monotonically from zero at the regular core to the asymptotic value  $Q(\rho) = \Lambda$ . Larger values of  $\mu$  localize the curvature more strongly and lead to a faster saturation of the enclosed topological charge. The horizontal dashed lines indicate representative fractions of the total charge associated with the discrete holonomy-invariance condition discussed in Sec. V.

where

$$\nabla_\mu = \partial_\mu - \Gamma_\mu \quad (6.2)$$

is the spinor covariant derivative and  $\Gamma_\mu$  denotes the spin connection induced by the geometry. For the conformal metric (2.1), a convenient orthonormal frame is given by

$$e^1 = e^{\Omega(r)} dr, \quad e^2 = e^{\Omega(r)} r d\theta. \quad (6.3)$$

Using the torsion-free Cartan structure equations, the corresponding spin connection can be obtained, and the Dirac equation can be written in Hamiltonian form. For stationary states, we can write the following *ansatz*:

$$\Psi(t, r, \theta) = e^{-iEt/\hbar} \Psi(r, \theta), \quad (6.4)$$

so that the effective Hamiltonian becomes

$$H = -i\hbar v_F e^{-\Omega(r)} \left[ \sigma^r \left( \partial_r + \frac{1}{2r} + \frac{1}{2} \Omega'(r) \right) + \frac{\sigma^\theta}{r} \partial_\theta \right], \quad (6.5)$$

where  $v_F$  is the Fermi velocity and  $\sigma^r$  and  $\sigma^\theta$  are the Pauli matrices defined in the local orthonormal frame. The term proportional to  $\Omega'(r)$  originates from the spin connection and describes the coupling between the Dirac pseudospin and the curvature distribution.

To separate the angular dependence, we write

$$\Psi_m(r, \theta) = \begin{pmatrix} f_m(r) e^{im\theta} \\ ig_m(r) e^{i(m+1)\theta} \end{pmatrix}, \quad m \in \mathbb{Z}. \quad (6.6)$$

We also define

$$k = \frac{E}{\hbar v_F}. \quad (6.7)$$

It is convenient to introduce the rescaled radial functions

$$f_m(r) = \frac{F_m(r)}{\sqrt{r}}, \quad g_m(r) = \frac{G_m(r)}{\sqrt{r}}, \quad (6.8)$$

which absorbs the contribution associated with the polar coordinate system. The Dirac equation then reduces to the coupled radial system

$$\left( \frac{d}{dr} + \frac{m+1}{r} + \frac{1}{2}\Omega'(r) \right) G_m(r) = k e^{\Omega(r)} F_m(r), \quad (6.9)$$

$$\left( \frac{d}{dr} - \frac{m}{r} + \frac{1}{2}\Omega'(r) \right) F_m(r) = -k e^{\Omega(r)} G_m(r). \quad (6.10)$$

These equations show that the conformal geometry affects the propagation of Dirac quasiparticles through two complementary mechanisms. The conformal factor  $e^{\Omega(r)}$  modifies the local momentum scale, whereas the spin connection introduces a geometric coupling directly associated with the curvature distribution. As discussed in the previous section, the same geometric structure also determines the holonomy and the effective topological charge enclosed by a given transport loop.

### B. Yukawa-Regularized Dirac Fermions

Specializing the general radial system to the Yukawa-regularized geometry, the conformal factor is given by Eq. (3.9), and the corresponding radial derivative becomes

$$\Omega'_Y(r) = -\frac{\Lambda}{2\pi r} + \frac{\Lambda\mu}{2\pi} K_1(\mu r), \quad (6.11)$$

where  $K_1$  denotes the modified Bessel function of the second kind of order one. Substituting these expressions into the radial Dirac equations yields

$$\left( \frac{d}{dr} + \frac{m+1}{r} - \frac{\Lambda}{4\pi r} + \frac{\Lambda\mu}{4\pi} K_1(\mu r) \right) G_m(r) = k e^{\Omega_Y(r)} F_m(r), \quad (6.12)$$

$$\left( \frac{d}{dr} - \frac{m}{r} - \frac{\Lambda}{4\pi r} + \frac{\Lambda\mu}{4\pi} K_1(\mu r) \right) F_m(r) = -k e^{\Omega_Y(r)} G_m(r). \quad (6.13)$$

These equations explicitly display the geometric contributions introduced by the Yukawa regularization. The terms proportional to  $K_1(\mu r)$  describe the finite-core corrections, while the  $1/r$  contributions determine the asymptotic behavior associated with the conical geometry.

### C. Asymptotic Regimes and Effective Angular Momentum

The radial Dirac equations obtained above allow a simple analysis of the quasiparticle dynamics in the limiting regions of the geometry. In the vicinity of the defect core, corresponding to  $r \ll \mu^{-1}$ , the modified Bessel function behaves as

$$K_1(\mu r) \simeq \frac{1}{\mu r} + \mathcal{O}(r \ln r). \quad (6.14)$$

Substituting this expansion into the radial equations, the singular terms cancel exactly,

$$-\frac{\Lambda}{4\pi r} + \frac{\Lambda\mu}{4\pi} K_1(\mu r) \simeq 0, \quad (6.15)$$

and one recovers

$$\left( \frac{d}{dr} + \frac{m+1}{r} \right) G_m(r) = k F_m(r), \quad (6.16)$$

$$\left( \frac{d}{dr} - \frac{m}{r} \right) F_m(r) = -k G_m(r). \quad (6.17)$$

Therefore, at sufficiently short distances, the quasiparticles behave as free massless Dirac fermions propagating in a locally flat environment.

In the opposite regime,

$$r \gg \mu^{-1}, \quad (6.18)$$

the modified Bessel function decays exponentially,

$$K_1(\mu r) \sim \sqrt{\frac{\pi}{2\mu r}} e^{-\mu r}, \quad (6.19)$$

and becomes negligible. The radial equations reduce to

$$\left( \frac{d}{dr} + \frac{m+1-\Lambda/4\pi}{r} \right) G_m(r) = k e^{\Omega_Y(r)} F_m(r), \quad (6.20)$$

$$\left( \frac{d}{dr} - \frac{m+\Lambda/4\pi}{r} \right) F_m(r) = -k e^{\Omega_Y(r)} G_m(r). \quad (6.21)$$

The effect of the geometry is therefore equivalent to a shift of the effective angular-momentum quantum number,

$$m \longrightarrow m + \frac{\Lambda}{4\pi}, \quad (6.22)$$

which is entirely induced by the spin connection. This shift plays a role analogous to an effective geometric flux, modifying the angular momentum quantum number through the spin connection. Such behavior is closely related to the geometric phases and holonomy effects associated with gravitational analogs of the Aharonov–Bohm effect [10]. This result indicates that the Yukawa regularization affects the quasiparticle dynamics only within a finite region of characteristic size  $\ell = \mu^{-1}$ . For  $r \gg \mu^{-1}$ , the finite-core corrections become exponentially suppressed, and the fermions recover the same effective angular-momentum shift and geometric phase associated with an ideal disclination. Thus, the regularization modifies the local structure of the defect while preserving its asymptotic topological signature.

#### D. Spinorial Holonomy and Geometric Phase

The angular momentum shift obtained in Eq. (6.22) and the spinorial holonomy derived below originate from the same spin-connection term appearing in the Dirac Hamiltonian (6.5). The former describes the asymptotic modification of the angular momentum quantum number, whereas the latter characterizes the geometric phase accumulated by the quasiparticles under parallel transport around the defect. The geometric effects encoded in the spin connection can also be interpreted in terms of the holonomy acquired by a Dirac spinor transported around the defect. The relevance of such geometric phases for graphene was emphasized in the pioneering work of Lammert and Crespi [19, 36], who showed that disclinations in graphitic cones induce nontrivial topological phases and effective gauge fluxes acting on low-energy Dirac quasiparticles. In this framework, the electronic properties depend not only on the local geometry but also on the global phase acquired when the quasiparticle encircles the defect.

For ideal graphitic cones, it was shown in Ref. [10] that the parallel transport of a spinor produces a geometric phase entirely determined by the deficit angle of the conical geometry. In that case, the corresponding spinorial holonomy operator may be written as

$$U_{\text{cone}} = \exp\left(-\frac{i}{2} \delta \sigma_3\right). \quad (6.23)$$

where  $\sigma_3$  acts on the pseudospin degrees of freedom. The holonomy associated with a general conformal metric was obtained in Ref. [11], where the deficit angle was shown to be directly related to the conformal factor through the spin connection. In the Yukawa-regularized geometry, the radius-dependent topological charge obtained in Sec. V yields the spinorial holonomy

$$U(\rho) = \exp\left[-\frac{i}{2} Q(\rho) \sigma_3\right]. \quad (6.24)$$

This expression shows that the geometric phase acquired by the spinor is directly controlled by the enclosed topological charge. Unlike the ideal conical case, where the phase is fixed by a single deficit angle, the Yukawa-regularized defect

generates a scale-dependent enclosed topological charge and the corresponding holonomy determined by the curvature enclosed by the transport loop.

The limiting cases follow directly from the behavior of  $Q(\rho)$ . Near the regular core,

$$\lim_{\rho \rightarrow 0} U(\rho) = \mathbb{I}, \quad (6.25)$$

indicating the absence of an accumulated geometric phase. In contrast,

$$\lim_{\rho \rightarrow \infty} U(\rho) = \exp\left(-\frac{i}{2} \Lambda \sigma_3\right). \quad (6.26)$$

which reproduces the spinorial holonomy of an ideal disclination. And thus, the Yukawa regularization preserves the asymptotic topological properties of the defect while modifying the way in which the geometric phase is accumulated. Rather than experiencing the entire topological charge at once, Dirac quasiparticles acquire the corresponding phase progressively as larger portions of the curvature distribution are enclosed by the transport path. The regularized core removes the singular interaction characteristic of ideal conical defects, while preserving the asymptotic topological signature through both the effective angular-momentum shift and the radius-dependent spinorial holonomy. This result provides a direct link between distributed curvature, scale-dependent topological charge, and the geometric phases acquired by graphene-like Dirac quasiparticles.

## VII. CONCLUSIONS

We presented an exact, closed-form construction of finite-core defects generated by Yukawa-type curvature distributions within a conformal cylindrical geometry. The solution regularizes the apex while preserving the global angle deficit determined by the total curvature, thereby bridging localized curvature near the core and conical behavior in the far field. The Yukawa screening introduces a finite interaction range, confining the geometric effects to the vicinity of the defect core while recovering the asymptotic conical geometry beyond a few screening lengths. This separation between the regularized core and the far-field regime provides a convenient setting for studying disclination-like defects in crystalline membranes and analog models of  $(2+1)$ -dimensional gravity. The exact solution obtained here provides a convenient starting point for analytical studies of parallel transport, geometric phases, and spinorial holonomy.

A central result of the present work is that the regularization does not alter the total topological charge of the defect, which remains fixed by the Gauss–Bonnet theorem, but changes the way in which this charge is distributed throughout space. Instead of being concentrated at a singular apex, the curvature is spread over a finite region, leading naturally to a radius-dependent topological charge and a corresponding scale-dependent holonomy. As a consequence, geometric phases are accumulated progressively as larger portions of the curvature distribution are enclosed, providing a direct connection between local geometric structure and global topological properties.

The extension to massless Dirac quasiparticles further shows that the regularized geometry affects not only the curvature profile but also the quantum dynamics. The singular interaction characteristic of ideal conical defects is replaced by a smooth finite-core structure, while the asymptotic topological signature survives through both the effective angular-momentum shift and the radius-dependent spinorial holonomy. These results provide a geometrically transparent description of how distributed curvature influences the behavior of graphene-like Dirac fermions.

Possible extensions include the study of wave and Dirac-particle scattering, bound states, geometric phases and transport phenomena in finite-core defect geometries. It would also be interesting to investigate torsional generalizations and more general curvature distributions, allowing a broader exploration of the interplay between geometry, topology, and quantum dynamics in condensed-matter and gravitational analog systems.

## DATA AVAILABILITY

No data are associated with this manuscript.

## DISCLAIMER

ChatGPT was used only as an auxiliary editorial tool for limited language refinement, text organization, formatting suggestions, and readability improvements. It also provided occasional assistance in checking the consistency and clarity of presentation. All scientific ideas, research design, literature analysis, mathematical derivations, physical interpretations, results, conclusions, and editorial decisions were developed and validated exclusively by the authors, who assume full responsibility for the content of this manuscript.

## ACKNOWLEDGMENTS

This work was supported by Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq) and Fundação de Apoio a Pesquisa do Estado da Paraíba (Fapesq-PB). G. Q. Garcia would like to thank Fapesq-PB for financial support (Grant BLD-ADT-A2377/2024). The work by C. Furtado is supported by the CNPq (project PQ Grant 1A No. 311781/2021-7). E. Brito would like to thank FAPESB (Fundação de Amparo à Pesquisa do Estado da Bahia) for financial support (Grant No APP0041/2023 and PPP0006/2024).

- 
- [1] K. Kondo, Proceedings of the 2nd Japan National Congress for Applied Mechanics , 41 (1952).
  - [2] B. A. Bilby, R. Bullough, and E. Smith, [Proceedings of the Royal Society A](#) **231**, 263 (1955).
  - [3] E. Kröner, Les Houches Session XXXV , 215 (1981).
  - [4] H. Kleinert, *Gauge Fields in Condensed Matter. Vol. II: Stresses and Defects* (World Scientific, Singapore, 1989).
  - [5] M. O. Katanaev and I. V. Volovich, [Annals of Physics](#) **216**, 1 (1992).
  - [6] M. O. Katanaev and I. V. Volovich, [Annals of Physics](#) **272**, 203 (1999).
  - [7] F. Moraes, [Brazilian Journal of Physics](#) **30**, 304 (2000).
  - [8] S. Fumeron, B. Berche, and F. Moraes, Journal of Geometry and Symmetry in Physics [10.1080/21680396.2022.2163515](#) (2023).
  - [9] S. Deser and R. Jackiw, [Annals of Physics](#) **153**, 405 (1984).
  - [10] C. Furtado, F. Moraes, and A. M. de M. Carvalho, [Physics Letters A](#) **372**, 5368 (2008).
  - [11] A. M. de M. Carvalho, C. A. de Lima Ribeiro, F. Moraes, and C. Furtado, [European Physical Journal Plus](#) **128**, 60 (2013).
  - [12] B. A. Klumov, JETP Letters **115**, 108 (2022).
  - [13] B. A. Klumov, JETP Letters **116**, 703 (2022).
  - [14] S. Carlip, [Nuclear Physics B](#) **324**, 106 (1989).
  - [15] J. González, F. Guinea, and M. A. H. Vozmediano, [Nuclear Physics B](#) **406**, 771 (1993).
  - [16] A. H. C. Neto, F. Guinea, N. M. R. Peres, K. S. Novoselov, and A. K. Geim, [Reviews of Modern Physics](#) **81**, 109 (2009).
  - [17] M. A. H. Vozmediano, M. I. Katsnelson, and F. Guinea, [Physics Reports](#) **496**, 109 (2010).
  - [18] A. Cortijo and M. A. H. Vozmediano, [Nuclear Physics B](#) **763**, 293 (2007).
  - [19] P. E. Lammert and V. H. Crespi, [Physical Review Letters](#) **85**, 5190 (2000).
  - [20] V. Volterra, in *Annales scientifiques de l'École normale supérieure*, Vol. 24 (1907) pp. 401–517.
  - [21] T. Eguchi, P. B. Gilkey, and A. J. Hanson, [Physics Reports](#) **66**, 213 (1980).
  - [22] A. M. de M. Carvalho, G. Q. Garcia, and C. Furtado, [International Journal of Modern Physics A](#) **41**, 2650045 (2026).
  - [23] M. Itagaki and C. A. Brebbia, [Engineering Analysis with Boundary Elements](#) **11**, 87 (1993).
  - [24] R. J. Duffin, Journal of Mathematical Analysis and Applications **35**, 105 (1971).
  - [25] M. Troyanov, Transactions of the American Mathematical Society **324**, 793 (1991).
  - [26] M. Abramowitz and I. A. Stegun, *Handbook of Mathematical Functions with Formulas, Graphs, and Mathematical Tables* (Dover Publications, New York, 1972).
  - [27] S. Deser, R. Jackiw, and G. t. Hooft, [Annals of Physics](#) **152**, 220 (1984).
  - [28] J. A. G. Vickers, [Classical and Quantum Gravity](#) **4**, 1 (1987).
  - [29] A. M. d. M. Carvalho and C. Furtado, [General Relativity and Gravitation](#) **39**, 1311 (2007).
  - [30] A. M. d. M. Carvalho, C. Furtado, and F. Moraes, [Journal of High Energy Physics](#) **2004**, 029 (2004).
  - [31] A. M. d. M. Carvalho, F. Moraes, and C. Furtado, [Classical and Quantum Gravity](#) **20**, 2063 (2003).
  - [32] K. Bakke, A. M. d. M. Carvalho, and C. Furtado, [European Physical Journal C](#) **63**, 149 (2009).
  - [33] L. C. Garcia de Andrade, A. M. d. M. Carvalho, and C. Furtado, [Europhysics Letters](#) **67**, 538 (2004).
  - [34] T. Rothman, G. F. R. Ellis, and J. Murugan, [Classical and Quantum Gravity](#) **18**, 1217 (2001).
  - [35] M. J. Bueno, G. Q. Garcia, A. M. de M. Carvalho, and C. Furtado, [Annals of Physics](#) **484**, 170182 (2025).
  - [36] P. E. Lammert and V. H. Crespi, [Physical Review B](#) **69**, 035406 (2004).